0816 EEG Meeting

Presenter: □□□ (Ian Chen)

Progress from the last 2 week:

1. Review FastICA and t-SNE paper line by line (Go through the mathematical derivation from the paper)

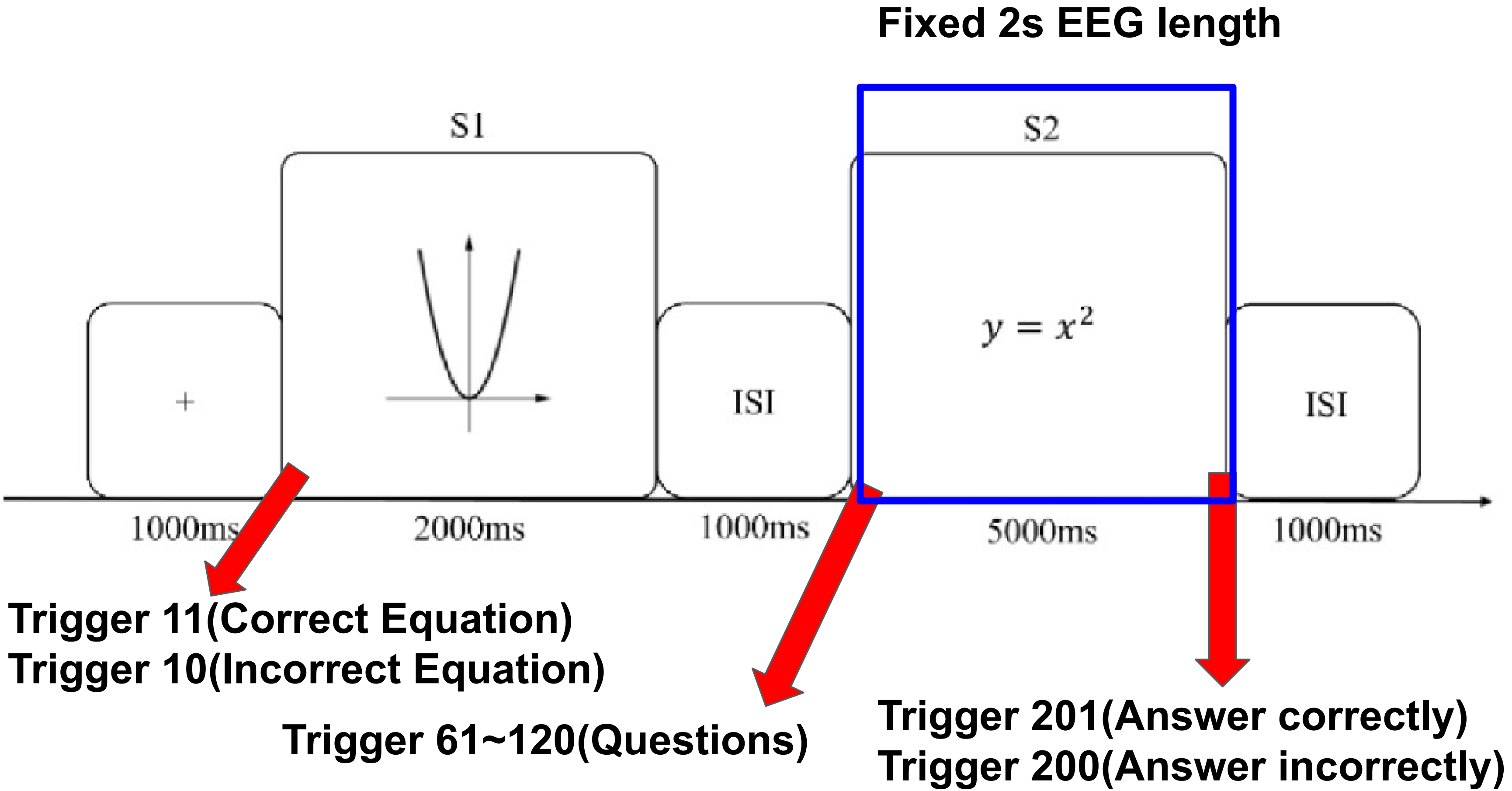
2. Work on IC clustering from the ICA encoder pipeline (Check the ICA alignment on regression performance)

3. Replacing the encoder with a full MLP architecture shows competitive results.

4. Try using EEGNet to encode components × time EEG data.

5. Investigate the memorization problem of the encoder

Experimental Protocol



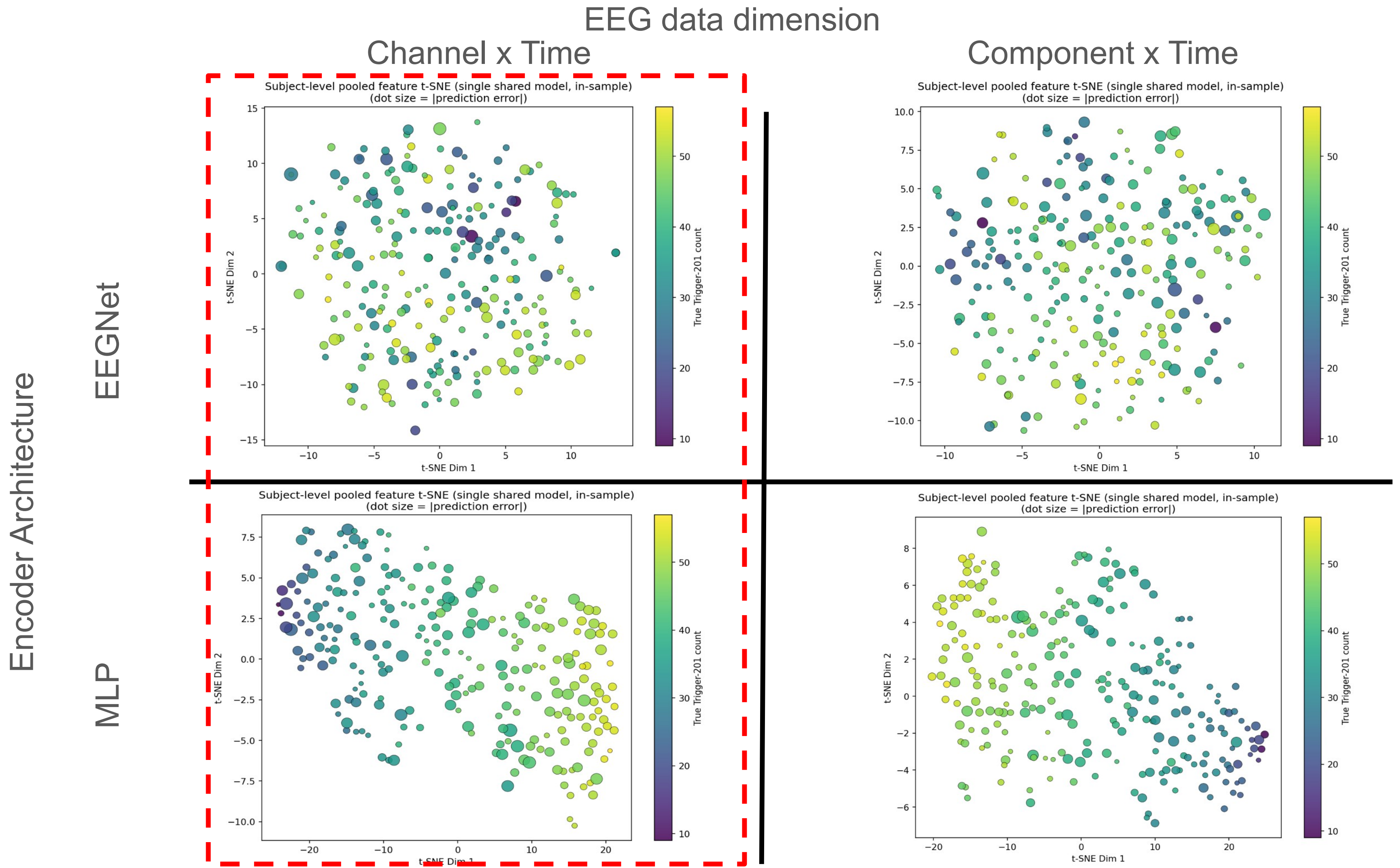
Preprocessing:

1. Notch filter at 60 Hz (line noise)
2. Bandpass 0.1–40 Hz
3. Average reference
4. ICA ocular artifact removal
5. Extracts −0.2 to 2.0 s epochs with a −0.2–0 s baseline correction
6. A rejection threshold of 150 μV peak-to-peak — auto-drops any epoch exceeding this. >> The number of epochs varies across subjects.
7. Each surviving epoch is tagged 1 (correct/201) or 0 (incorrect/200)

Resampled to 128 Hz post-epoching.

Validation: GroupKFold-by-subject

With ~256 subjects and N_SPLITS=5, each fold holds out roughly 1/5 of subjects (~51) as test, trains on the rest.


EEG data dimension
Channel x Time
Component x Time
Encoder Architecture
EEGNet
MLP
Subject-level pooled feature t-SNE (single shared model, in-sample)
(dot size = |prediction error|)
t-SNE Dim 1
t-SNE Dim 2
True Trigger-201 count